

Standard CNN Models

VGGNet and GoogLeNet

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Motivation and Historical Context for VGGNet

- After AlexNet, researchers hypothesized that **increasing depth** (stacking more convolutional layers) would improve the hierarchical feature representation, allowing the network to capture complex patterns better.
- However, computational resources and training stability were challenges.
- The VGG models were introduced by Karen Simonyan and Andrew Zisserman from Oxford University in their 2014 paper titled “*Very Deep Convolutional Networks for Large-Scale Image Recognition*”.
- The primary objective was to investigate the effect of increasing the depth of CNNs on large-scale image recognition tasks.

Ref:

Simonyan, Karen and Andrew Zisserman. “Very Deep Convolutional Networks for Large-Scale Image Recognition.” *CoRR* abs/1409.1556 (2014)

VGG Architecture

Innovations in VGG architecture

- ✓ Instead of large filters (11×11 in AlexNet) VGG used stacked small 3×3 filters.
- ✓ Reduced number of parameters compared to large filters.
- ✓ Simple architecture

The VGG family includes various configurations with different depths, denoted by the letter "VGG" followed by the number of weight layers.

Variations include VGG-11, VGG-13, VGG-16, VGG-19

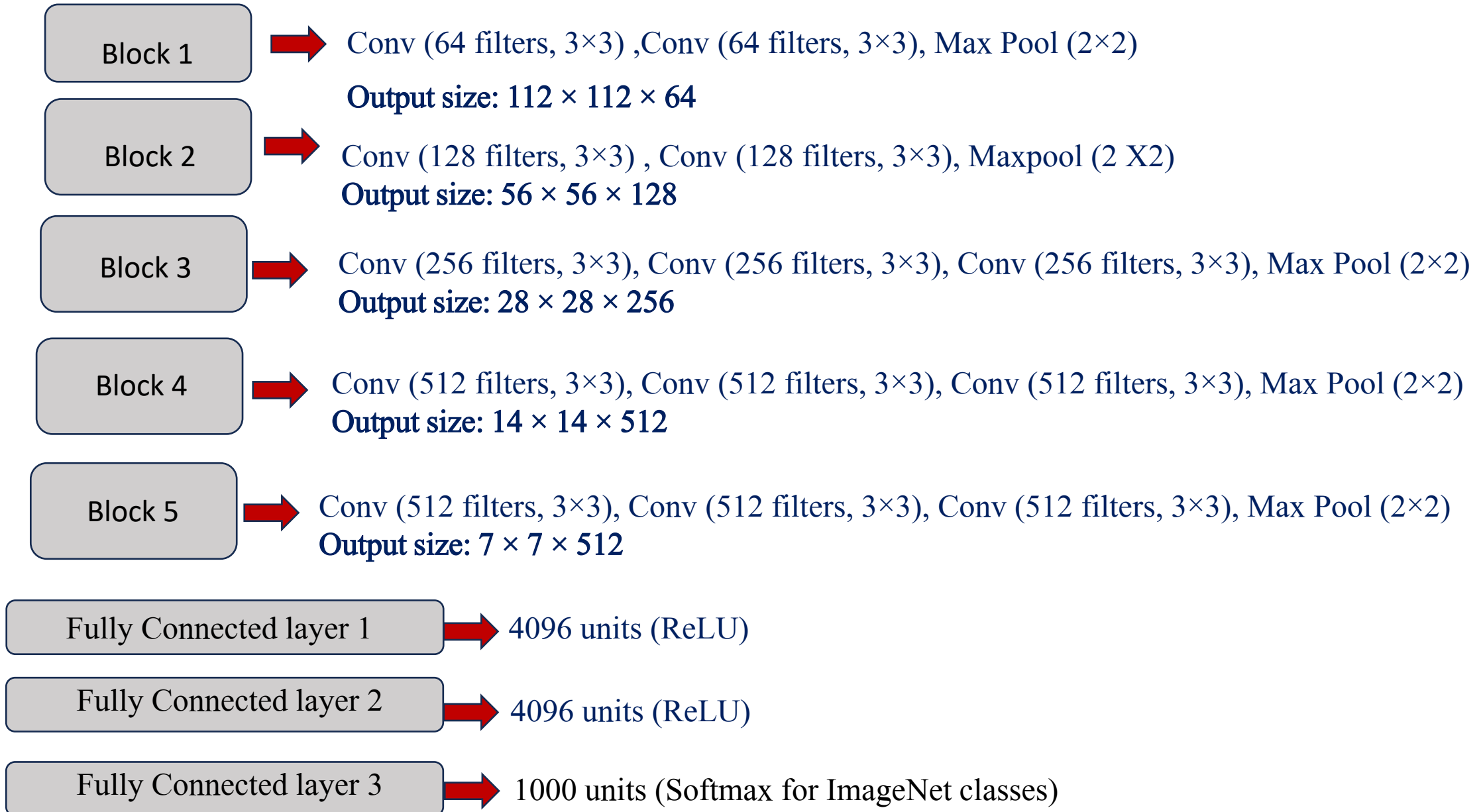
VGG-16

- 13 convolutional layers
- 3 fully connected layers
- 138M parameters

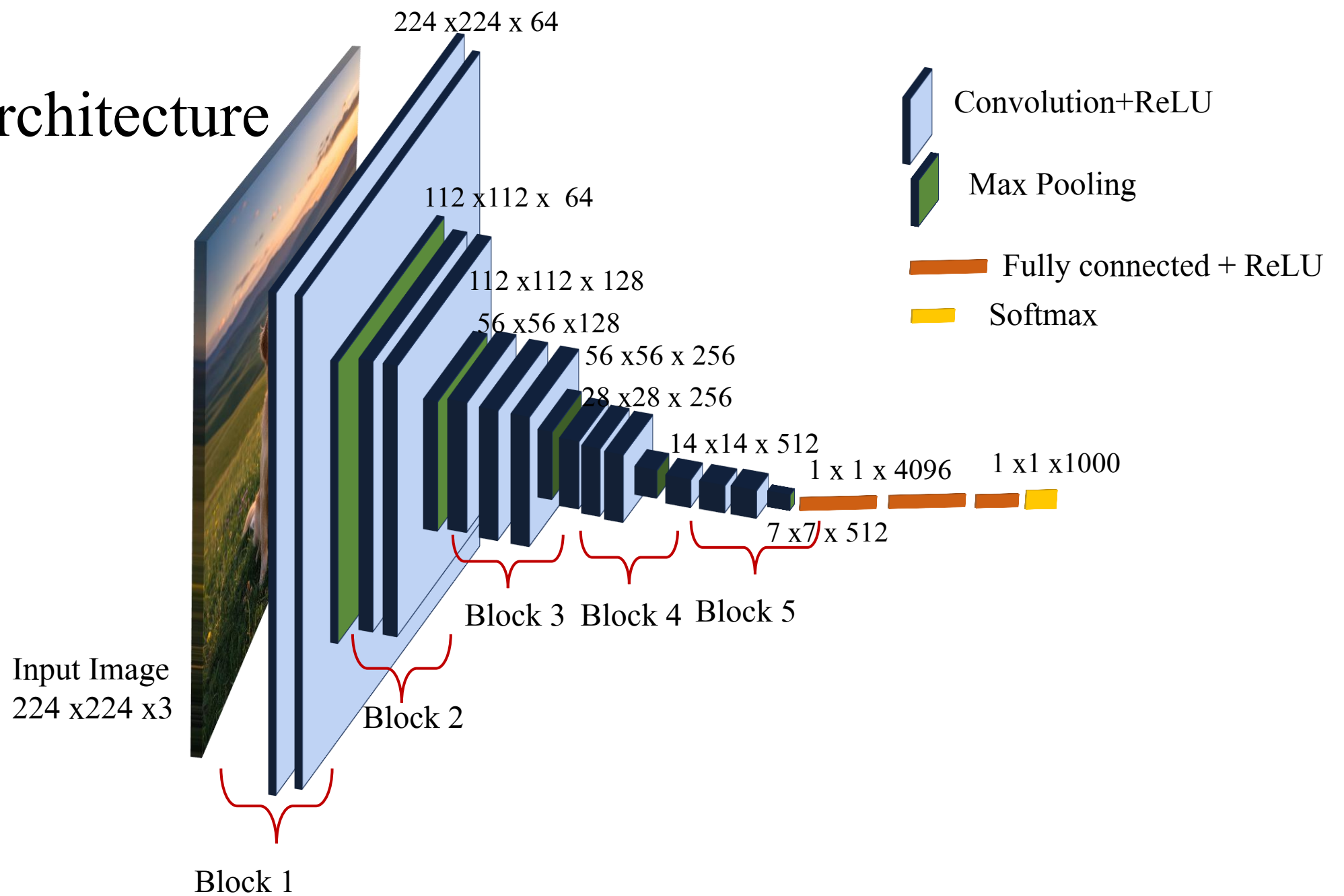
VGG-19

- 16 convolutional layers
- 3 fully connected layers
- 144 M parameters

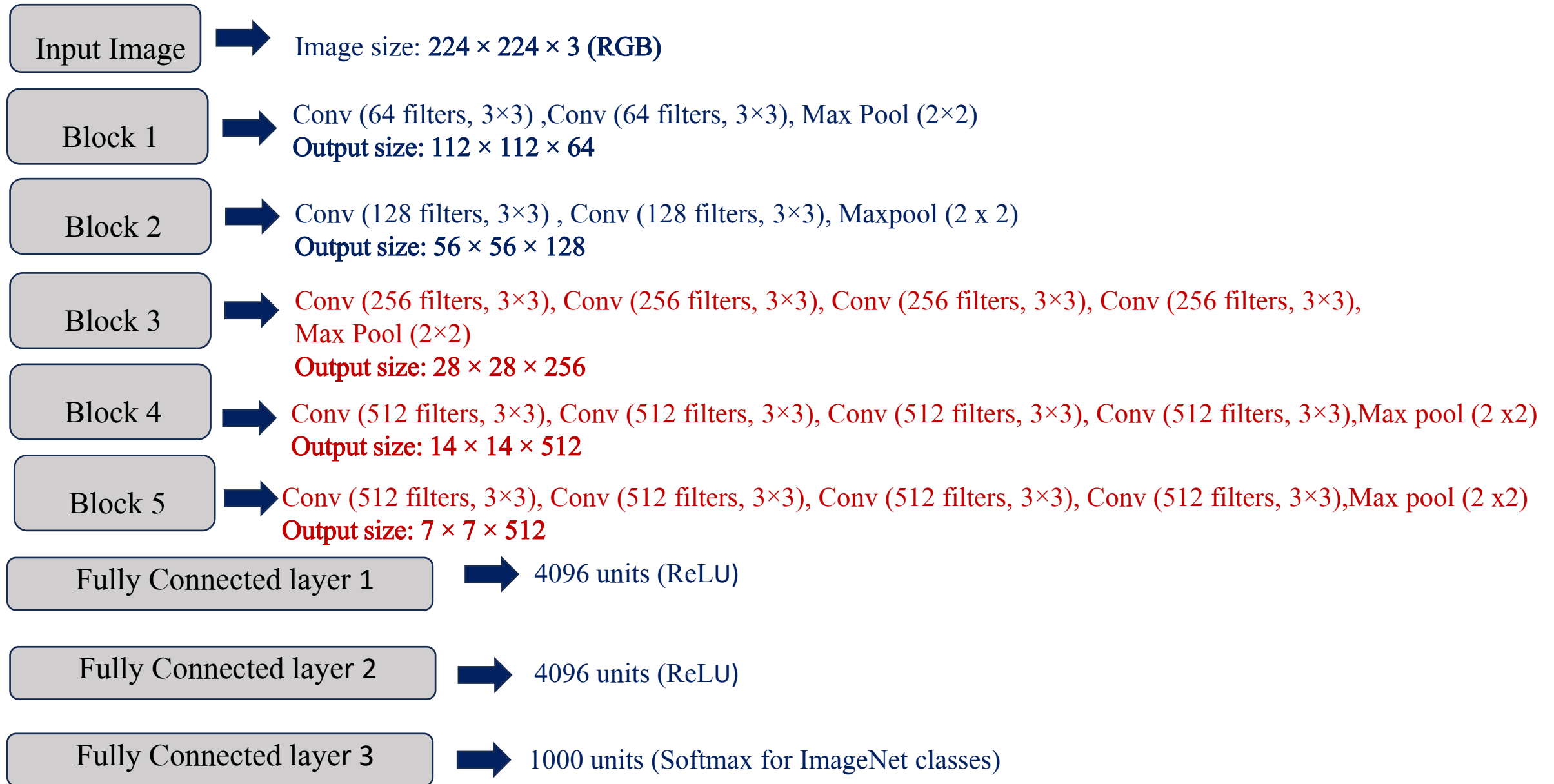
Detailed Architecture of VGG-16



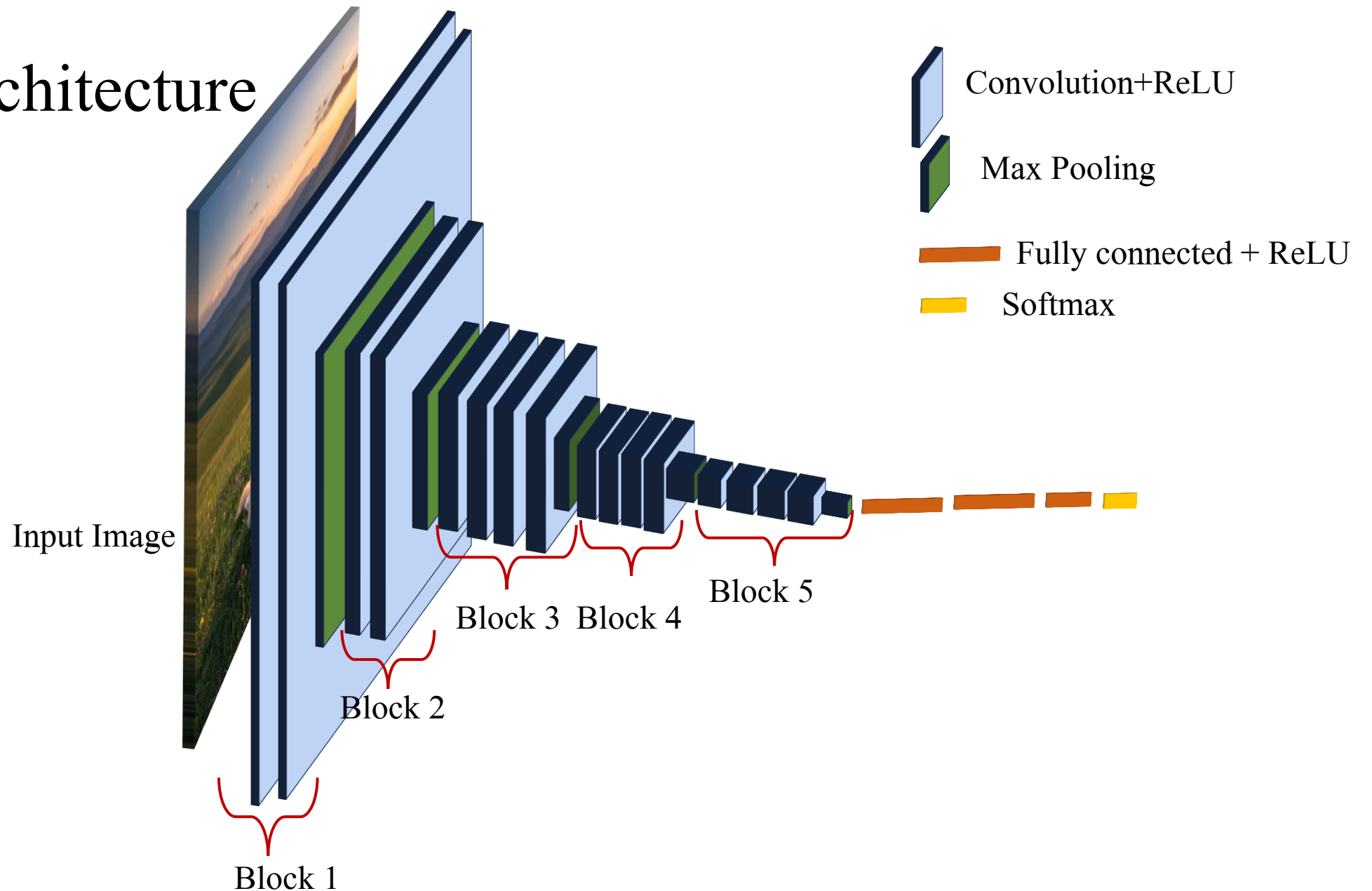
VGG 16 Architecture



VGG 19 Architecture



VGG 19 Architecture



VGG19 is **deeper**, but the accuracy gain over VGG16 is **tiny** ($\sim 0.2\%$ in ImageNet Dataset)

Impact and Limitations of VGGNet

VGG16 secured the 1st runner-up of the ILSVRC (ImageNet Large Scale Visual Recognition Competition) 2014 in the classification task with a top-5 classification error of 7.32%.

It is the state of the art object recognition model

Limitations of VGG 16:

Huge Number of Parameters

VGG16 has **~138 million parameters** (most in the fully connected layers).
Requires large storage (over 500 MB for the model).
It is very slow to train.

High Computational Cost

Needs powerful GPUs and lots of training time.

GoogLeNet

GoogLeNet (Inception V1)

Motivation

- Previous CNN architectures such as AlexNet and VGGNet had a limited number of layers. (shallow architecture)
- This limited their ability to learn complex patterns in data.
- Architectures were not very computationally efficient.

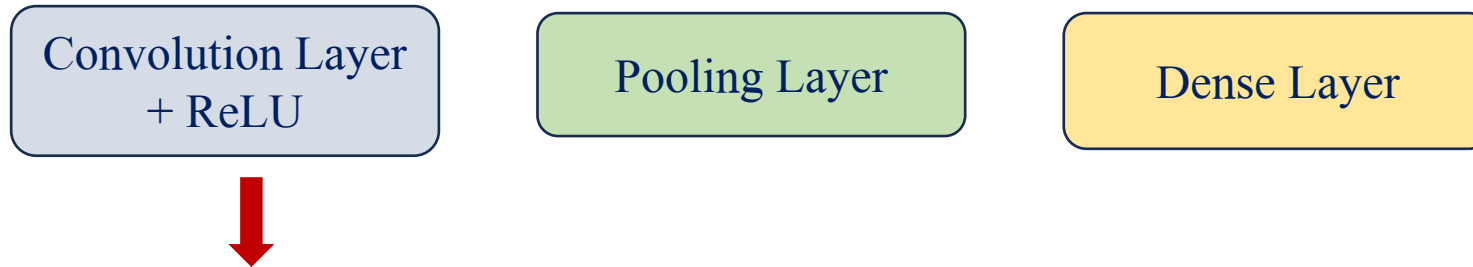
- GoogLeNet addressed the challenges of previous CNN architectures by introducing the concept of **inception modules**.
- It is a 22-layer deep convolutional network

Unveiled by researchers at Google in 2014, it introduces a novel approach to designing deep network which achieved

- Exceptionally accuracy
- Computationally efficient
- **Won the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2014.**

Understanding Inception Module

Standard CNN modules are made up of



Traditional CNNs use *one* kernel size per layer — like a 3×3 or 5×5 convolution.

But features in an image can vary in **scale** (edges, textures, objects of different sizes).

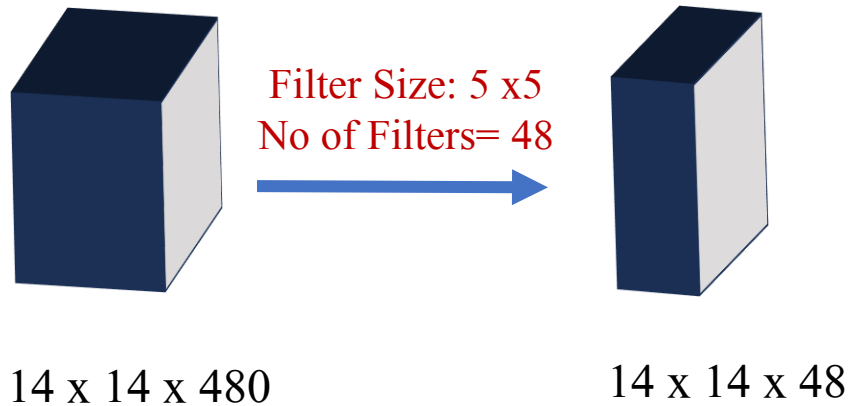
Core idea of
Inception Module



- ✓ *Why not look at the image using multiple receptive fields at the same time?*
- ✓ Instead of choosing one type of convolution filter (like 3×3 or 5×5), Inception module applies multiple filters in parallel to the same input and concatenates their outputs.

Dimension Reduction Using 1 x1 Convolution Module

Traditional 5×5 convolution operation

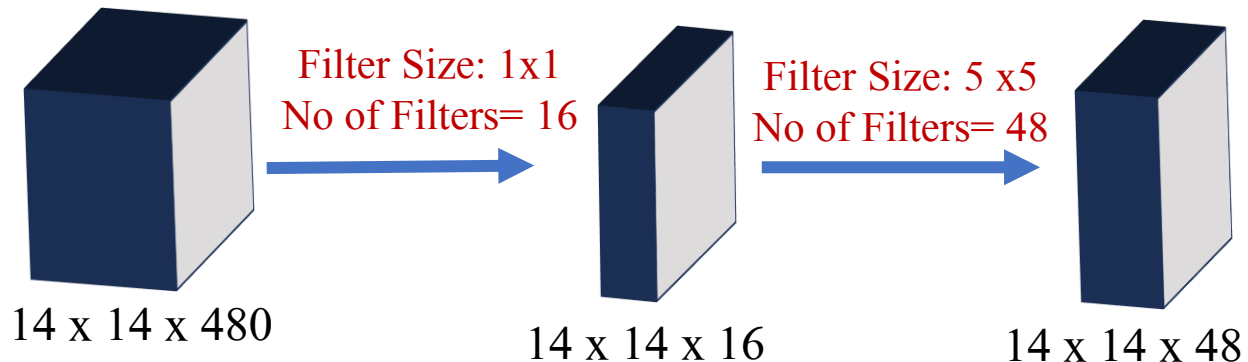


No of operations:

$$(H_{\text{out}} \times W_{\text{out}} \times C_{\text{out}}) \times (F_{\text{width}} \times F_{\text{height}}) \times (\text{no. of input channels})$$

Number of operations involved here is $(14 \times 14 \times 48) \times (5 \times 5 \times 480) = 112.9\text{M}$

5×5 convolution using 1×1 convolution



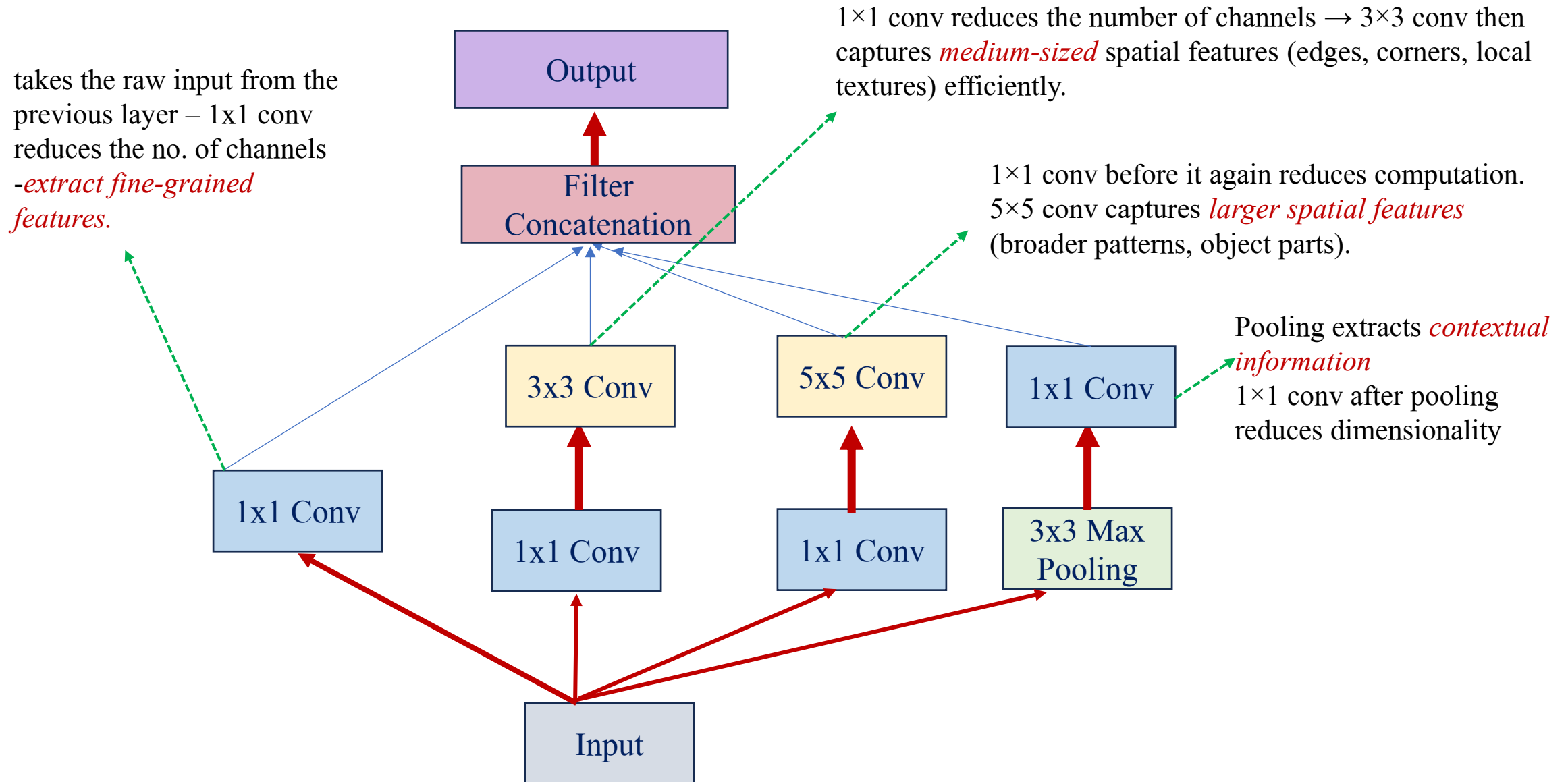
Number of operations for 1×1 convolution = $(14 \times 14 \times 16) \times (1 \times 1 \times 480) = 1.5\text{M}$

Number of operations for 5×5 convolution = $(14 \times 14 \times 48) \times (5 \times 5 \times 16) = 3.8\text{M}$

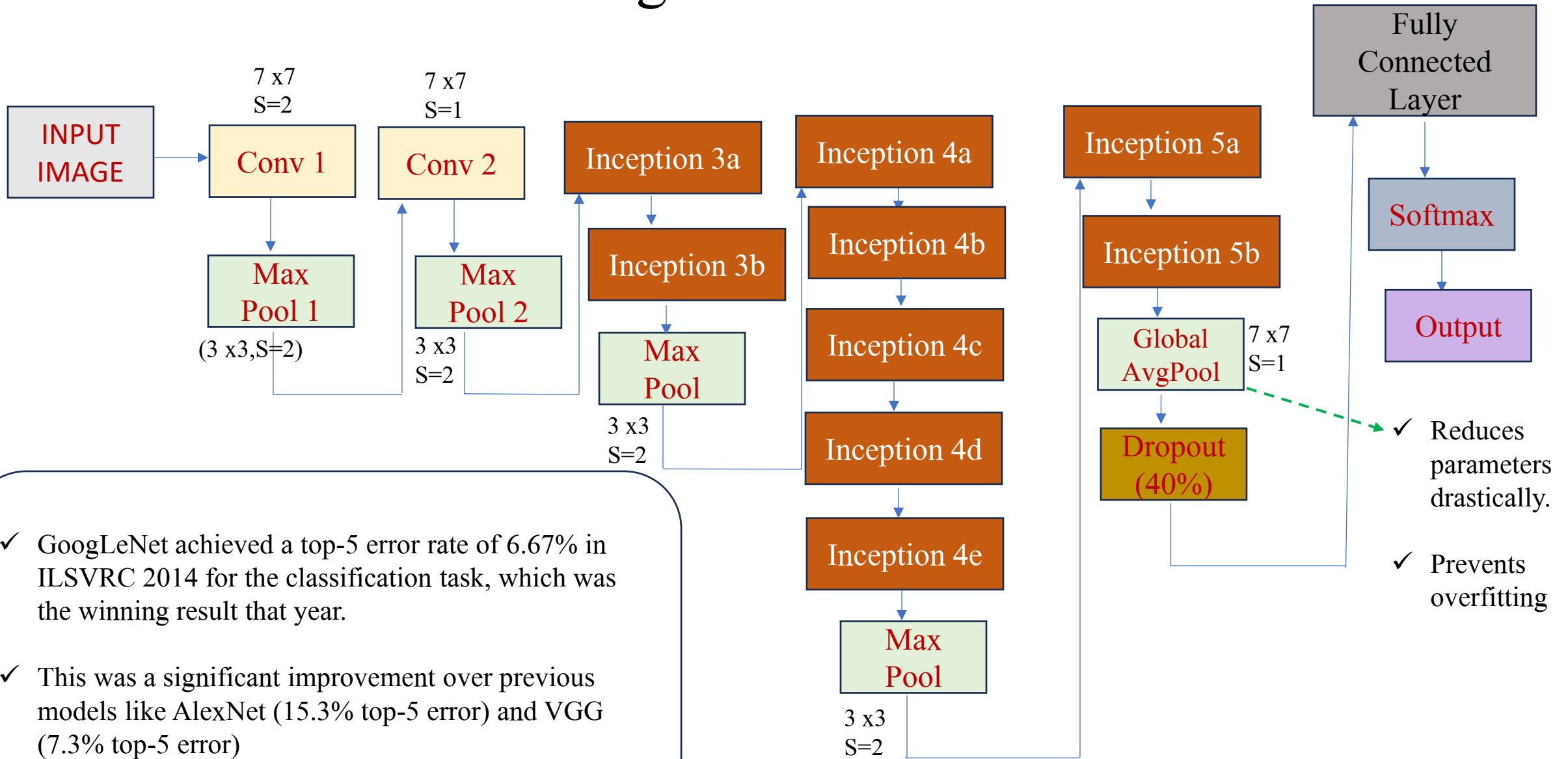
Total: $1.5\text{M} + 3.8\text{M} = 5.3\text{M}$

1×1 convolution helps in dimensionality reduction which reduces the computation

Inception Module



GoogLeNet Architecture



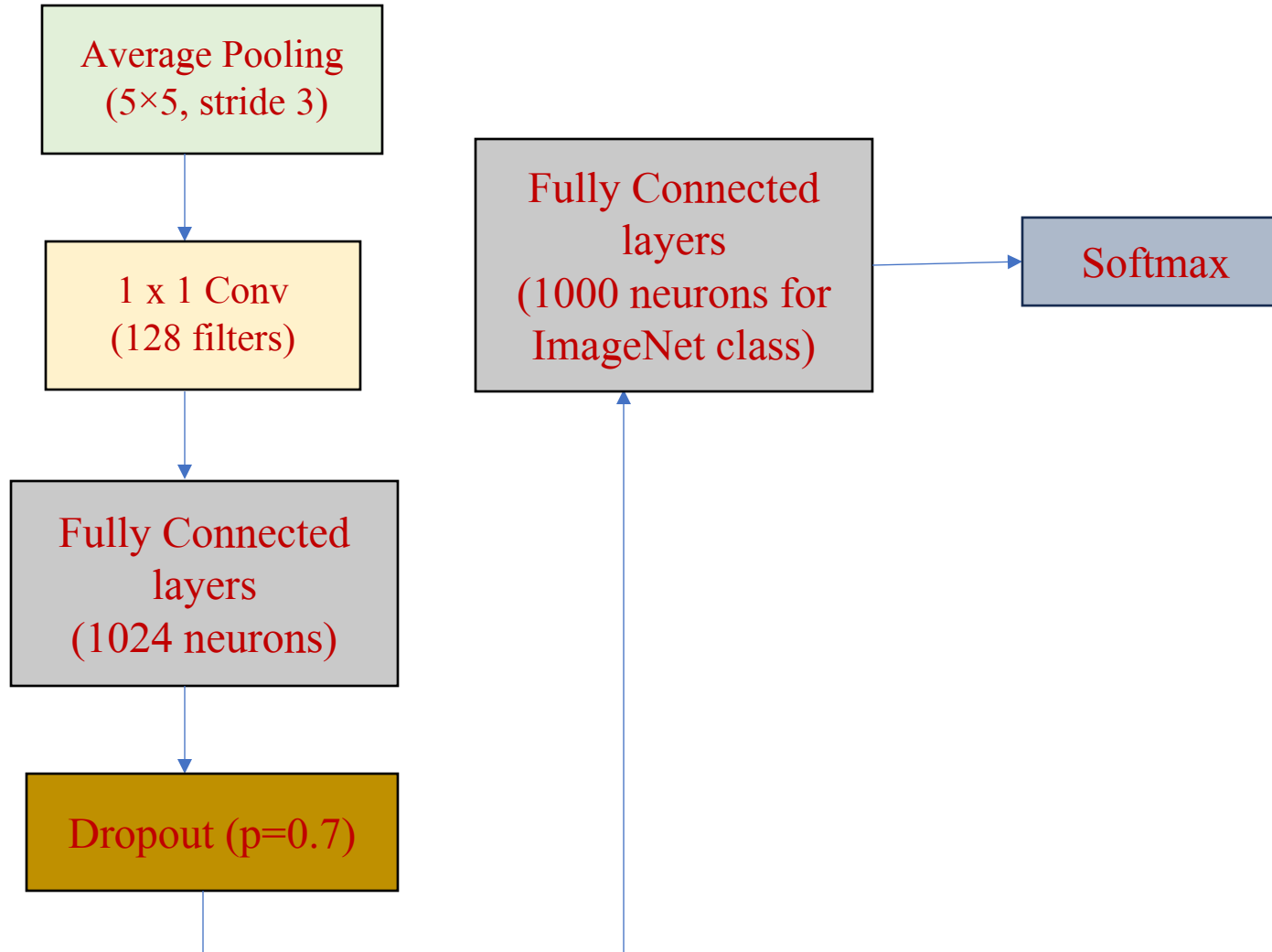
- ✓ GoogLeNet achieved a top-5 error rate of 6.67% in ILSVRC 2014 for the classification task, which was the winning result that year.
- ✓ This was a significant improvement over previous models like AlexNet (15.3% top-5 error) and VGG (7.3% top-5 error)

- GoogLeNet has 22 layers, so during training the gradient signal weakens as it moves backward — this is the vanishing gradient problem.
- To mitigate this problem, two smaller classifiers have been added to the model.
- These help in propagating gradients back to earlier layers, improving training stability.

These are called Auxiliary Classifiers

- One of them is placed after Inception module 4a
- Other one after Inception module 4d
- They branch out from the main network.

Auxiliary Classifier for Training GoogLeNet



Variants of GoogLeNet

✓ Inception v2 (2015)

- Inception v2 included improvements such as batch normalization and shortcut connections.
- Refined the inception modules by replacing larger convolutions with smaller, more efficient ones.
- These changes improved accuracy and reduced training time.

✓ Inception v3 (2015)

The v3 model refined Inception v2 by using atrous convolution (dilated convolutions that expand the network's receptive field without sacrificing resolution and significantly increasing network parameters).

Summary

Next Session

ResNet Model

Thank You